

Therefore the leading 1's of B must be on the diagonal and because B is the reduced row-echelon form of A we also know that all the entries above and below the leading 1's must be zeroes. This however, is exactly I_n . Therefore, if B does not have a row of all zeroes in it then we must have that $B = I_n$.

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Matrix Arithmetic & Operations

One of the biggest impediments that some people have in learning about matrices for the first time is trying to take everything that they know about arithmetic of real numbers and translate that over to matrices. As you will eventually see much of what you know about arithmetic of real numbers will also be true here, but there is also a few ideas/facts that will no longer hold here. To make matters worse there are some rules of arithmetic of real numbers that will work occasionally with matrices but won't work in general. So, keep this in mind as you go through the next couple of sections and don't be too surprised when something doesn't quite work out as you expect it to.

This section is devoted mostly to developing the arithmetic of matrices as well as introducing as well as introducing a couple of operations on matrices that don't really have an equivalent operation in real numbers. We will see some of the differences between arithmetic of real numbers and matrices mentioned above in this section. We will also see more of them in the next section when we delve into the properties of matrix arithmetic in more detail.

Okay, let's start off matrix arithmetic by defining just what we mean when we say that two matrices are equal.

Definition 1 If A and B are both $n \times m$ matrices then we say that $A = B$ provided corresponding entries from each matrix are equal. Or in other words, $A = B$ provided $a_{ij} = b_{ij}$ for all i and j .

Matrices of different sizes cannot be equal.

Example 1 Consider the following matrices.

$$A = \begin{bmatrix} -9 & 123 \\ 3 & -7 \end{bmatrix} \quad B = \begin{bmatrix} -9 & b \\ 3 & -7 \end{bmatrix} \quad C = \begin{bmatrix} -9 \\ 3 \end{bmatrix}$$

For these matrices we have that $A \neq C$ and $B \neq C$ since they are different sizes and so can't be equal. The fact that C is essentially the first column of both A and B is not important to determining equality in this case. The size of the two matrices is the first thing we should look at in determining equality.

Next, $A = B$ provided we have $b = 123$. If $b \neq 123$ then we will have $A \neq B$.

Next we need to move on to addition and subtraction of two matrices.

Definition 2 If A and B are both $n \times m$ matrices then $A \pm B$ is a new $n \times m$ matrix that is found by adding/subtracting corresponding entries from each matrix. Or in other words,

$$A \pm B = [a_{ij} \pm b_{ij}]$$

Matrices of different sizes cannot be added or subtracted.

Example 2 For the following matrices perform the indicated operation, if possible.

$$A = \begin{bmatrix} 2 & 0 & -3 & 2 \\ -1 & 8 & 10 & -5 \end{bmatrix} \quad B = \begin{bmatrix} 0 & -4 & -7 & 2 \\ 12 & 3 & 7 & 9 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 0 & 2 \\ -4 & 9 & 5 \\ 6 & 0 & -6 \end{bmatrix}$$

(a) $A + B$

(b) $B - A$

(c) $A + C$

Solution

(a) Both A and B are the same size and so we know the addition can be done in this case. Once we know the addition can be done there really isn't all that much to do here other than to just add the corresponding entries here to get the results.

$$A + B = \begin{bmatrix} 2 & -4 & -10 & 4 \\ 11 & 11 & 17 & 4 \end{bmatrix}$$

(b) Again, since A and B are the same size we can do the difference and as like the previous part there really isn't all that much to do. All that we need to be careful with is the order. Just like with real number arithmetic $B - A$ is different from $A - B$. So, in this case we'll subtract the entries of A from the entries of B .

$$B - A = \begin{bmatrix} -2 & -4 & -4 & 0 \\ 13 & -5 & -3 & 14 \end{bmatrix}$$

(c) In this case because A and C are different sizes the addition can't be done. Likewise, $A - C$, $C - A$, $B + C$, $C - B$, and $B - C$ can't be done for the same reason.

We now need to move into multiplication involving matrices. However, there are actually two kinds of multiplication to look at : Scalar Multiplication and Matrix Multiplication. Let's start with scalar multiplication.

Definition 3 If A is any matrix and c is any number then the product (or **scalar multiple**), cA , is a new matrix of the same size as A and it's entries are found by multiplying the original entries of A by c . In other words $cA = [c a_{ij}]$ for all i and j .

Note that in the field of Linear Algebra a number is often called a **scalar** and hence the name scalar multiple since we are multiplying a matrix by a scalar (number). From this point on we will generally call numbers scalars.

Before doing an example we need to get another quick definition out of the way. If $A_1, A_2, \dots, A_n$ are all matrices of the same size and $c_1, c_2, \dots, c_n$ are scalars then the **linear combination** of $A_1, A_2, \dots, A_n$ with **coefficients** $c_1, c_2, \dots, c_n$ is,

$$c_1 A_1 + c_2 A_2 + \dots + c_n A_n$$

This may seem like a silly thing to define but we'll be using linear combination in quite a few places in this class and so we need to get used to seeing them.

Example 3 Given the matrices

$$A = \begin{bmatrix} 0 & 9 \\ 2 & -3 \\ -1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 8 & 1 \\ -7 & 0 \\ 4 & -1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 3 \\ -2 & 5 \\ 10 & -6 \end{bmatrix}$$

compute $3A + 2B - \frac{1}{2}C$.

Solution

So, we're really being asked to compute a linear combination here. We'll do that by first computing the scalar multiplies and then performing the addition and subtraction. Note as well that in the case of the third scalar multiple we are going to consider the scalar to be a positive $\frac{1}{2}$ and leave the minus sign out in front of the matrix. Here is the work for this problem.

$$3A + 2B - \frac{1}{2}C = \begin{bmatrix} 0 & 27 \\ 6 & -9 \\ -3 & 3 \end{bmatrix} + \begin{bmatrix} 16 & 2 \\ -14 & 0 \\ 8 & -2 \end{bmatrix} - \begin{bmatrix} 1 & \frac{3}{2} \\ -1 & \frac{5}{2} \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 15 & \frac{55}{2} \\ -7 & -\frac{23}{2} \\ 0 & 4 \end{bmatrix}$$

We now need to move into matrix multiplication, however before we do the general case let's look at a special case first since this will help with the general case.

Suppose that we have the following two matrices,

$$\mathbf{a} = \begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

So, $\mathbf{a}$ is a row matrix and $\mathbf{b}$ is a column matrix and they have the same number of entries. Then the product of $\mathbf{a}$ and $\mathbf{b}$ is defined to be,

$$\mathbf{ab} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n$$

It is important to note that this product can only be done if $\mathbf{a}$ and $\mathbf{b}$ have the same number of entries. If they have a different number of entries then this product is not defined.

Example 4 Compute $\mathbf{ab}$ given that,

$$\mathbf{a} = \begin{bmatrix} 4 & -10 & 3 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} -4 \\ 3 \\ 8 \end{bmatrix}$$

Solution

There is not really a whole lot to do here other than use the definition given above.

$$\mathbf{ab} = (4)(-4) + (-10)(3) + (3)(8) = -22$$

Now let's move onto general matrix multiplication.

Definition 4 If A is an $n \times p$ matrix and B is a $p \times m$ matrix then the product (or **matrix multiplication**) is a new matrix with size $n \times m$ whose ij^{th} entry is found by multiplying row i of A times column j of B .

So, just like with addition and subtraction, we need to be careful with the sizes of the two matrices we're dealing with. However, with multiplication we need to be a little more careful. This definition tells us that the product AB is only defined if A (i.e. the first matrix listed in the product) has the same number of columns as B (i.e. the second matrix listed in the product) has rows. If the number of columns of the first matrix listed is not the same as the number of rows of the second matrix listed then the product is not defined.

An easy way to check that a product is defined is to write down the two matrices in the order that we want to multiply them and underneath them write down the sizes as shown below.

$$\begin{array}{ccc} A & B & = & AB \\ n \times p & p \times m & & n \times m \end{array}$$

If the two inner numbers are equal then the product is defined and the size of the product will be given by the outside numbers.

Example 5 Compute AC and CA for the following two matrices, if possible.

$$A = \begin{bmatrix} 1 & -3 & 0 & 4 \\ -2 & 5 & -8 & 9 \end{bmatrix} \quad C = \begin{bmatrix} 8 & 5 & 3 \\ -3 & 10 & 2 \\ 2 & 0 & -4 \\ -1 & -7 & 5 \end{bmatrix}$$

Solution

Okay, let's first do AC . Here are the sizes for A and C .

$$\begin{array}{ccc} A & C & = & AC \\ 2 \times 4 & 4 \times 3 & & 2 \times 3 \end{array}$$

So, the two inner numbers (4 and 4) are the same and so the multiplication can be done and we can see that the new size of the matrix is 2×3 . Now, let's actually do the multiplication. We'll go through the first couple of entries in the product in detail and

then do the remaining entries a little quicker.

To get the number in the first row and first column of AC we'll multiply the first row of A by the first column of B as follows,

$$(1)(8) + (-3)(-3) + (0)(2) + (4)(-1) = 13$$

If we next want the entry in the first row and second column of AC we'll multiply the first row of A by the second column of B as follows,

$$(1)(5) + (-3)(10) + (0)(0) + (4)(-7) = -53$$

Okay, at this point, let's stop and insert these into the product so we can make sure that we've got our bearings. Here's the product so far,

$$\begin{bmatrix} 1 & -3 & 0 & 4 \\ -2 & 5 & -8 & 9 \end{bmatrix} \begin{bmatrix} 8 & 5 & 3 \\ -3 & 10 & 2 \\ 2 & 0 & -4 \\ -1 & -7 & 5 \end{bmatrix} = \begin{bmatrix} 13 & -53 & \square \\ \square & \square & \square \end{bmatrix}$$

As we can see we've got four entries left to compute. For these we'll give the row and column multiplications but leave it to you to make sure we used the correct row/column and put the result in the correct place. Here's the remaining work.

$$(1)(3) + (-3)(2) + (0)(-4) + (4)(5) = 17$$

$$(-2)(8) + (5)(-3) + (-8)(2) + (9)(-1) = -56$$

$$(-2)(5) + (5)(10) + (-8)(0) + (9)(-7) = -23$$

$$(-2)(3) + (5)(2) + (-8)(-4) + (9)(5) = 81$$

Here is the completed product.

$$\begin{bmatrix} 1 & -3 & 0 & 4 \\ -2 & 5 & -8 & 9 \end{bmatrix} \begin{bmatrix} 8 & 5 & 3 \\ -3 & 10 & 2 \\ 2 & 0 & -4 \\ -1 & -7 & 5 \end{bmatrix} = \begin{bmatrix} 13 & -53 & 17 \\ -56 & -23 & 81 \end{bmatrix}$$

Now let's do CA . Here are the sizes for this product.

$$\begin{array}{ccc} C & A & = \quad CA \\ 4 \times 3 & 2 \times 4 & \quad N/A \end{array}$$

Okay, in this case the two inner numbers (3 and 2) are NOT the same and so this product can't be done.

So, with this example we've now run across the first real difference between real number arithmetic and matrix arithmetic. When dealing with real numbers the order in which we write a product doesn't affect the actual result. For instance $(2)(3)=6$ and $(3)(2)=6$. We can flip the order and we get the same answer. With matrices however, we will have to

be very careful and pay attention to the order in which the product is written down. As this example has shown the product AC could be computed while the product CA is not defined.

Now, do not take the previous example and assume that all products will work that way. It is possible for both AC and CA to be defined as we'll see in the next example.

Example 6 Compute BD and DB for the given matrices, if possible.

$$B = \begin{bmatrix} 3 & -1 & 7 \\ 10 & 1 & -8 \\ -5 & 2 & 4 \end{bmatrix} \quad D = \begin{bmatrix} -1 & 4 & 9 \\ 6 & 2 & -1 \\ 7 & 4 & 7 \end{bmatrix}$$

Solution

First, notice that both of these matrices are 3×3 matrices and so both BD and DB are defined. Again, it's worth pointing out that this example differs from the previous example in that both the products are defined in this example rather than only one being defined as in the previous example. Also note that in both cases the product will be a new 3×3 matrix.

In this example we're going to leave the work of verifying the products to you. It is good practice so you should try and verify at least one of the following products.

$$BD = \begin{bmatrix} 3 & -1 & 7 \\ 10 & 1 & -8 \\ -5 & 2 & 4 \end{bmatrix} \begin{bmatrix} -1 & 4 & 9 \\ 6 & 2 & -1 \\ 7 & 4 & 7 \end{bmatrix} = \begin{bmatrix} 40 & 38 & 77 \\ -60 & 10 & 33 \\ 45 & 0 & -19 \end{bmatrix}$$

$$DB = \begin{bmatrix} -1 & 4 & 9 \\ 6 & 2 & -1 \\ 7 & 4 & 7 \end{bmatrix} \begin{bmatrix} 3 & -1 & 7 \\ 10 & 1 & -8 \\ -5 & 2 & 4 \end{bmatrix} = \begin{bmatrix} -8 & 23 & -3 \\ 43 & -6 & 22 \\ 26 & 11 & 45 \end{bmatrix}$$

This example leads us to yet another difference (although it's related to the first) between real number arithmetic and matrix arithmetic. In this example both BD and DB were defined. Notice however that the products were definitely not the same. There is nothing wrong with this so don't get excited about it when it does happen. Note however that this doesn't mean that the two products will never be the same. It is possible for them to be the same and we'll see at least one case where the two products are the same in a couple of sections.

For the sake of completeness if A is an $n \times p$ matrix and B is a $p \times m$ matrix then the entry in the i^{th} row and j^{th} column of AB is given by the following formula,

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ip}b_{pj}$$

This formula can be useful on occasion, but is really used mostly in proofs and computer programs that compute the product of matrices.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

If we now denote the coefficient matrix by A , the column matrix containing the unknowns by $\mathbf{x}$ and the column matrix containing the b_i 's by $\mathbf{b}$, we can write the system in the following **matrix form**,

$$A\mathbf{x} = \mathbf{b}$$

In many of the section to follow we'll write general systems of equations as $A\mathbf{x} = \mathbf{b}$ given its compact nature in order to save space.

Now that we've gotten the basics of matrix arithmetic out of the way we need to introduce a couple of matrix operations that don't really have any equivalent operations with real numbers.

Definition 5 If A is an $n \times m$ matrix then the **transpose** of A , denoted by A^T , is an $m \times n$ matrix that is obtained by interchanging the rows and columns of A . So, the first row of A^T is the first column of A , the second row of A^T is the second column of A , etc. Likewise, the first column of A^T is the first row of A , the second column of A^T is the second row of A , etc.

On occasion you'll see the transpose defined as follows,

$$A = [a_{ij}]_{n \times m} \quad \Rightarrow \quad A^T = [a_{ji}]_{m \times n} \quad \text{for all } i \text{ and } j$$

Notice the difference in the subscripts. Under this definition, the entry in the i^{th} row and j^{th} column of A will be in the j^{th} row and i^{th} column of A^T .

Notice that these two definitions are really the same definition, they just don't look like they are the same at first glance.

Definition 6 If A is a square matrix of size $n \times n$ then the **trace** of A , denoted by $\text{tr}(A)$, is the sum of the entries on main diagonal. Or,

$$\text{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}$$

If A is not square then the trace is not defined.

Example 10 Determine the transpose and trace (if it is defined) for each of the following matrices.

$$A = \begin{bmatrix} 4 & 10 & -7 & 0 \\ 5 & -1 & 3 & -2 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 2 & -6 \\ -9 & 1 & -7 \\ 5 & 0 & 12 \end{bmatrix} \quad C = \begin{bmatrix} 9 \\ -1 \\ 8 \end{bmatrix}$$

$$D = [15] \quad E = \begin{bmatrix} -12 & -7 \\ -7 & 10 \end{bmatrix}$$

Solution

There really isn't all that much to do here other than to go through the definitions. Note as well that the trace will only not be defined for A and C since these matrices are not square.

$$A^T = \begin{bmatrix} 4 & 5 \\ 10 & -1 \\ -7 & 3 \\ 0 & -2 \end{bmatrix} \quad \text{tr}(A): \text{Not defined since } A \text{ is not square.}$$

$$B^T = \begin{bmatrix} 3 & -9 & 5 \\ 2 & 1 & 0 \\ -6 & -7 & 12 \end{bmatrix} \quad \text{tr}(B) = 3 + 1 + 12 = 16$$

$$C^T = [9 \quad -1 \quad 8] \quad \text{tr}(C): \text{Not defined since } C \text{ is not square.}$$

$$D^T = [15] \quad \text{tr}(D) = 15$$

$$E^T = \begin{bmatrix} -12 & -7 \\ -7 & 10 \end{bmatrix} \quad \text{tr}(E) = -12 + 10 = -2$$

In the previous example note that $D^T = D$ and that $E^T = E$. In these cases the matrix is called **symmetric**. So, in the previous example D and E are symmetric while A , B , and C , are not symmetric.

Properties of Matrix Arithmetic and the Transpose

In this section we're going to take a quick look at some of the properties of matrix arithmetic and of the transpose of a matrix. As mentioned in the previous section most of the basic rules of real number arithmetic are still valid in matrix arithmetic. However, there are a few that are no longer valid in matrix arithmetic as we'll be seeing.

We've already seen one of the real number properties that doesn't hold in matrix arithmetic. If a and b are two real numbers then we know by the commutative law for